

AI: Artificial Intelligence

Symbolic AI → Expert System (Specific Problems)

KB → Inference Engine

Fuzzy logic → Image classification.

Rules X Data ✓
(Learn)

Deep Learning → Subset of ML. Algorithms are diff. Inspired by our brains. (Storilises after 1 point)

Perception

ML → You have to provide features. (Performance increases with more data)

DL → Features are detected by system (Performance increases with more data)

↳ powerful technology over big data

FL → Don't know what features.

③ - Types of ML

Supervised Unsupervised Semisupervised Reinforcement (Amount of supervision needed)

→ Regression
→ Classification
→ Clustering
→ Dimensionality redⁿ
→ Anomaly detection
→ Association

Supervised → IQ, CGPA, Y/N process } 5000 students

input			output	
iq	cgpa	Y/N		
87	7.1	Y		
111	8.9	Y		
75	6.3	N		

input ✓
output ✓ → output

(i) Regression → Output is numerical, A number

(ii) Classification → Output is classification
Y/N/NO, character/categor

Data
→ Numerical
→ Categorical

Age, sex, cgpa, iq
Gender, Nation

ML is a field of CS that uses statistical techniques to give computer systems the ability to "learn" with data, without being explicitly programmed.

- Data Gathering
- Hardware.

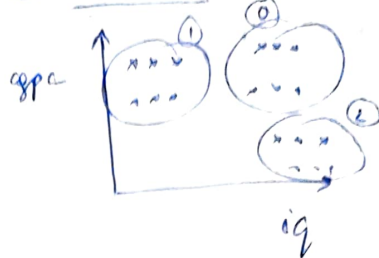
Unsupervised →

- Clustering
- Dimensionality Redⁿ
- Association
- Anomaly detection

5000 students

iq	cgpa	
70	7.0	0
60	6.0	1
100	9	2

① Clustering



Clustering is used widely in industries.

(ii) Dimensionality

Redⁿ →

1000 input columns.

Removes these extra columns

No. of rooms	No. of washrooms
	sq. ft. area

PCA Algo

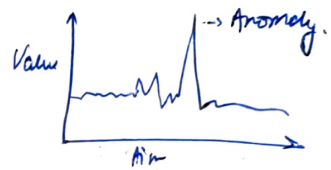
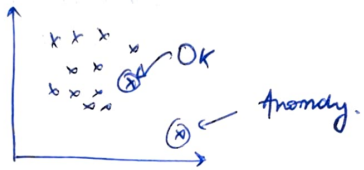
feature extraction

Visualising MNIST (Visualization)

→ image of handwritten numbers.

(iii)

Anomaly detection



(iv) Association
rule
learning

→ E.g. Selling beer with baby diapers.
hidden correlation.

Semisupervised →

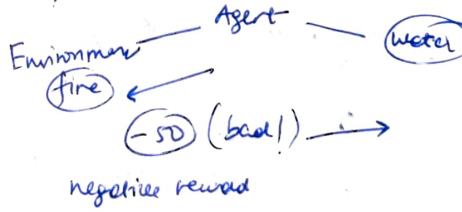
Manual labeling & human intervention.

Label / Name

1 data/photo & rest all data will be used.

Reinforcement
(No data)

Self

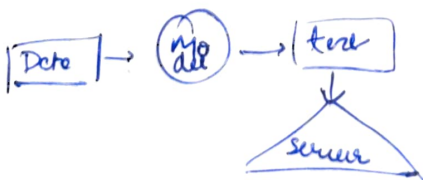


- 1 Observe
- 2 Selection action policy
- 3 Action
- 4 Get reward
- 5 Update policy
- 6 Iterate until optimal path found.

Via- ④ - Batch Machine Learning

i) Batch vs Online ML
(Offline)

Batch → Conventional way with 1 lot.

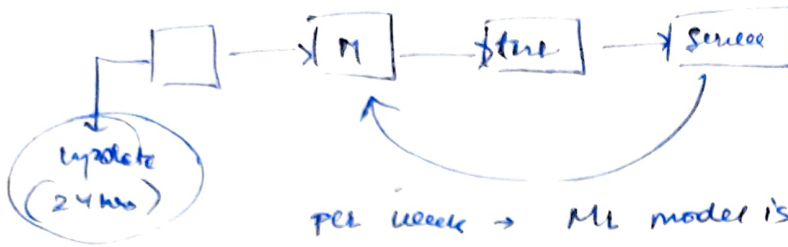


Production - server on which website is running

Coding on machine → Development environment

Problem →

- 1) Static ✓
- 2) NO new data is used to train ✓

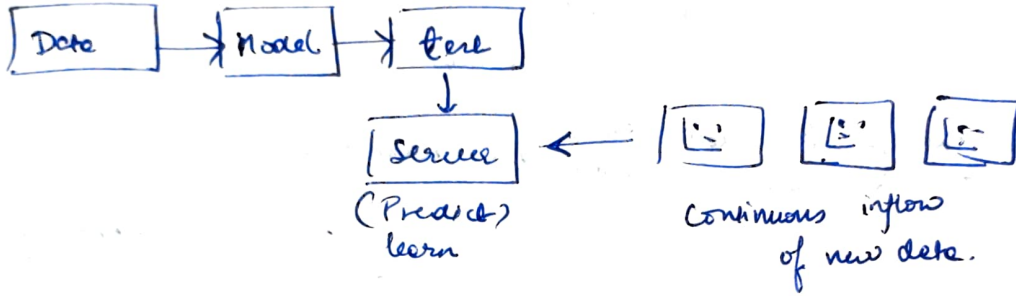


Train model with entire data & again deploy it on server.

per week $\rightarrow$ ML model is trained with entire data.

Disadvantages $\rightarrow$ 1. Lots of Data ; 2. Hardware Limitation 3. Availability

⑤ Online Machine Learning



E.g., Chabors.

Mini-batches of data

Q. When to use?

- (i) Concept drift / Volatile Scenarios
- (ii) Faster solution.
- (iii) Cost-Effective

Review

Vowpal Wabbit

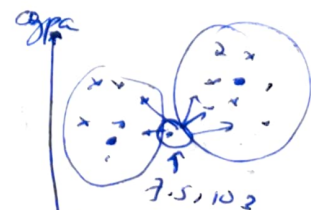
④. Learning Rate 5. Out of Core Learning $\rightarrow$ Data is huge so you can't load in 1 go. We need to take it in batches.

Disadvantages $\rightarrow$ 1. Tricky to Use 2. Risky. (Active monitoring system)

Offline	Online
$\rightarrow$ Image Classification $\rightarrow$ Scikit, TensorFlow, Pytorch, keras, Spark ML $\rightarrow$ Easier to implement	$\rightarrow$ Finance, economics, health. - Active research, $\rightarrow$ Scikit - multiflow $\rightarrow$ Difficult to implement & manage.

⑥ Instance Based Vs Model Based Learning

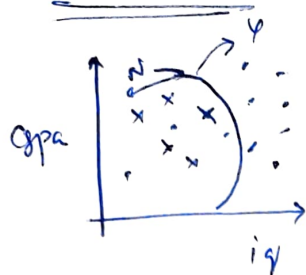
Learning	(i) Instance Based	$\rightarrow$ NO training or learning
memorizing	iq	cgpc
	80	8
generalising	70	7
		placement
		Y
		N



knn, RPF, Kernel functions

No 2 $\rightarrow$ nearest number knn

(ii) Model Based



Most ML Algos use this approach.



Model

- Predict using model
- Can throw away I/O data
- Requires a known model form
- Less storage is needed
- Predict an unscore scaling invariant
- Predict an unscore mode

⑦ Challenges in ML

i) Data collection

API Web scraping

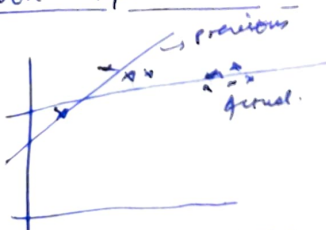
ii) Insufficient Data / Labeled Data

A
(100)
 M_1

B
(10⁶)
 M_2 ✓

Unreasonable effectiveness of data.

iii) Non Representative data



Sampling Noise
Sampling Bias

iv) Poor Quality of Data →

60% time is used to improve quality of data.

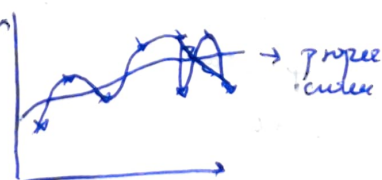
v) Irrelevant Features →

Garbage In → Garbage Out

Fitness →

Age | wt | height | location → irrelevant
↓
BMI → feature engineering

vi) Overfitting →



vii) Underfitting

